



The relationship between organizational focus on AI, financial growth and sustainable development: Evidence from Europe

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ABSTRACT

This study examines the link between organizations' focus on AI and their environmental, social, and governance (ESG) score. Furthermore, this study examines the relationship between organizations' AI focus and financial performance, measured by return on assets (ROA) and Tobin's Q. This manuscript relies on observations from a balanced panel of data comprising 432 publicly listed companies headquartered in Europe. The sample excludes banks and insurance companies, given their distinct accounting, governance, and capital structure standards. The sample consists of observations spanning from 2015 to 2023. Observations are gathered from LSEG Data & Analytics. We conduct baseline regression models. To ensure rigor, we also applied Hausman tests, variance inflation factors (VIF), and several robustness checks. The present manuscript is grounded in the economic theory framework. The empirical findings indicate: I) a positive and significant association between organizations' AI focus and their environmental ($b = 0.127^{***}$; $p = 0.001$) and social pillar scores ($b = 0.072^{**}$; $p = 0.023$); II) a positive and significant link with financial performance (ROA: $b = 0.094^{**}$; $p = 0.012$; TobinQ: 0.103^{*} ; $p = 0.051$) and; III) a positive but statistically insignificant relationship with governance pillar scores ($b = 0.030$; $p = 0.166$). The obtained results yield significant contributions to both theory and practice. Specifically, the obtained results clarify and reconcile previously heterogeneous findings in the literature. Furthermore, it emphasizes that an organizational focus on AI may contribute to advancing the United Nations Sustainable Development Goals, while simultaneously enhancing financial performance.

1. Introduction

Artificial intelligence (AI) has recently gained momentum as a powerful tool that can deeply transform how organizations operate, pursue strategy and innovate. AI is expected to significantly impact organizations, markets and economies worldwide (Agrawal et al., 2019a, 2019b; Appio et al., 2024; Li et al., 2025).

Generally speaking, AI is detailed as an information technology (IT) that allows machines to perform generative and cognitive functions. These include tasks traditionally associated with understanding, learning, and interacting (Feuerriegel et al., 2023; Gupta et al., 2024). Although research on AI has been ongoing for decades, its popularity has surged only recently. The growing interest in AI is largely driven by the increasing computational power. This improvement results from

advances in digitalization, computational tools, and data processing capabilities (Gill et al., 2022; Xiao et al., 2025). The integration of digital systems, connecting machines, communication devices, and sensors, combined with the exponential growth of digitized data, has created new opportunities for organizations. These opportunities allow them to leverage insights through innovative applications aimed at improving operational efficiency (Corazza et al., 2024; Frasca et al., 2024; Gupta et al., 2020; Lanzolla et al., 2020; Leone et al., 2021; La Torre et al., 2023). In fact, research estimates suggest that AI could generate an economic impact surpassing several trillion dollars by 2030. This potential stems from its ability to process vast amounts of unstructured data and generate predictive insights (Mishra et al., 2022). AI rapidly gained significance among businesses all over the globe due to its recognized ability to shape future entities' competitiveness (Enholm

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